

Investigating Cross-Attention Fusion for Deepfake Image Detection Utilizing ViT–CNN Architectures

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Abstract

The growing realism of synthetic facial alteration necessitates detection methods that can manage distribution shifts between datasets. Convolutional networks adeptly capture nuanced spatial features, whereas transformer designs excel in representing long-range semantic connections. This research presents an adaptive token-level fusion architecture that combines convolutional feature maps with transformer embeddings via a unidirectional cross-attention method. The suggested approach facilitates dynamic interaction between local texture descriptors and global contextual representations, rather than relying on traditional feature concatenation, enabling the model to selectively enhance discriminative cues. A progressive three-stage fine-tuning technique is employed to enhance optimization stability during the adaptation of pretrained backbones. The system is trained on a combined OpenForensics and FaceForensics++ dataset and assessed under both intra- and cross-dataset conditions. Experimental findings indicate exceptional in-domain performance and increased robustness upon domain shift, attaining 99.25% AUC in the intra-dataset context and 76.20% AUC on Celeb-DF (V2), hence affirming superior generalization ability.

Keywords: *Deepfake Image Forensics, Cross-Attention Fusion, CNN–Transformer Hybrid, Token-Level Feature Interaction, Cross-Dataset Generalization.*

I. INTRODUCTION

The rapid advancement of generative models, particularly Generative Adversarial Networks (GANs) and diffusion-based models, has enabled the creation of highly realistic synthetic images, commonly referred to as deepfakes. While these technologies have been widely adopted in entertainment and digital content creation, they also pose significant risks to society, including misinformation dissemination, identity fraud, and violations of personal privacy. The increasing visual realism of deepfake images makes it extremely difficult for humans to distinguish manipulated content from authentic images, thereby undermining

trust in visual media and motivating extensive research on automatic deepfake image detection techniques [1, 2].

Convolutional neural networks (CNNs) have been the dominant approach for deepfake image detection in recent years. By exploiting local receptive fields, CNN-based methods are effective at learning texture inconsistencies and low-level artifacts introduced during image manipulation [3]. Many state-of-the-art CNN detectors achieve strong performance under intra-dataset evaluation settings, where training and testing samples share similar data distributions. However, their performance often degrades significantly when applied to unseen datasets or novel manipulation techniques [4]. This limited generalization capability is largely attributed to the strong local inductive bias of CNNs, which restricts their ability to capture long-range dependencies and global contextual relationships within images.

To address the limitations of CNN-only approaches, Vision Transformers (ViTs) have recently emerged as a promising alternative for visual recognition tasks. ViTs represent images as sequences of patches and employ self-attention mechanisms to model global dependencies across the entire image [5]. This global modeling capability allows transformer-based models to capture holistic representations that are less sensitive to local artifact variations. Recent studies have demonstrated that Vision Transformer-based approaches can achieve improved robustness and generalization in deepfake detection, particularly in cross-dataset evaluation scenarios. As a result, ViTs have attracted increasing attention as an effective direction for overcoming the generalization challenges inherent in CNN-based deepfake detection methods.

To overcome the limitations of static or shallow feature fusion strategies (e.g., feature-level fusion without explicit inter-branch attention in ViXNet [6]), we propose a Cross-Modal Attention ViT–CNN Hybrid Network (CMAViT) that dynamically integrates complementary strengths of CNNs and Vision Transformers (ViTs) via cross-modal attention mechanisms.

CNNs (e.g., Xception) are known to be effective at modeling local spatial features, making them suitable for capturing fine-grained visual artifacts in manipulated images [2, 3], but struggle with long-range dependencies and global inconsistencies. ViTs, conversely, capture global contextual correlations effectively through self-attention [5], yet may overlook subtle local details [7].

By combining these modalities, the hybrid approach yields a richer representation: local fine-grained manipulation traces from CNNs inform global inconsistencies from ViTs, improving robustness and generalization [6].

The key advancement is the use of cross-modal attention for fusion: CNN-derived local features serve as keys/values, while ViT patch embeddings act as queries (or bidirectional), enabling adaptive, context-aware interaction between local and global representations. This dynamic fusion outperforms static concatenation by better exploiting inter-modal relationships, reducing redundancy, and enhancing detection of subtle, diverse deepfake artifacts, as demonstrated in cross-attention fusion approaches for vision tasks [8].

II. RELATED WORKS

A. CNN-based Deepfake Image Detection

Convolutional Neural Networks (CNNs) have been widely applied to deepfake image detection due to their strong capability in learning local visual patterns and texture-level artifacts. Early CNN-based approaches focused on detecting spatial inconsistencies introduced during face manipulation, such as unnatural textures and blending errors. Among various architectures, XceptionNet has become a standard backbone for deepfake detection because of its efficiency and strong feature representation ability [3].

Rössler et al. introduced the FaceForensics++ benchmark and demonstrated that CNN-based models can achieve high detection accuracy when trained and evaluated on the same manipulation methods [9]. However, Li et al. later showed that the performance of CNN-based detectors drops significantly on high-quality and more realistic deepfakes provided by the Celeb-DF dataset, highlighting the limited generalization ability of CNNs across different domains [2]. These findings indicate that CNN-based methods heavily rely on dataset-specific artifacts and struggle to handle unseen manipulation techniques.

B. Transformer-based Deepfake Detection

To address the limitations of CNN-based approaches, Transformer-based models have recently been explored for deepfake detection. Vision

Transformers (ViTs) model images as sequences of fixed-size patches and employ self-attention mechanisms to capture long-range dependencies and global contextual information [5]. This property allows Transformers to detect holistic facial inconsistencies rather than relying solely on local texture cues.

Recent studies have shown that Transformer-based architectures, such as Swin Transformer, improve robustness and generalization performance in cross-dataset deepfake detection scenarios by effectively balancing local and global feature modeling [10]. Although Transformer-based methods generally require more computational resources and larger datasets, their ability to capture global relationships makes them a promising direction for real-world deepfake detection.

C. Hybrid CNN-Transformer Architecture

Hybrid CNN–Transformer architectures leverage the complementary strengths of Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) for deepfake image detection. CNNs, such as Xception [3], excel at capturing local spatial features and fine-grained manipulation artifacts (e.g., texture inconsistencies and blending boundaries) due to their inductive biases of locality and translation invariance [2, 3]. However, they struggle with long-range dependencies and global semantic inconsistencies, leading to limited generalization across datasets [4].

In contrast, ViTs [5] model images as patch sequences and use self-attention to capture global contextual relationships, offering improved robustness to local variations and better performance in cross-dataset scenarios. Pure ViT approaches may overlook subtle local forgery traces without sufficient local inductive bias [7].

ViXNet [6] represents a key hybrid model in this domain, combining a ViT backbone with an Xception CNN branch for deepfake image and video forgery detection. It achieves competitive intra-dataset performance by integrating local artifact detection from CNN with global reasoning from ViT. Despite this, ViXNet relies on relatively shallow or static fusion, which limits explicit interaction between branches and hinders optimal exploitation of complementary cues, resulting in suboptimal generalization under distribution shifts.

The proposed CMAViT advances this by introducing explicit cross-modal attention, enabling dynamic fusion between CNN feature maps and ViT token embeddings. This hybrid design outperforms ViXNet-style static hybrids by better integrating local fine-grained traces with global inconsistencies, leading to enhanced robustness and cross-dataset performance.

D. Feature Fusion Strategies

Feature fusion is critical in hybrid CNN–Transformer models to combine local and global representations effectively. Conventional strategies, such as simple concatenation, element-wise summation, or late fusion (as in ViXNet [6]), provide basic integration but lack adaptive interaction. These static methods often introduce redundancy, fail to selectively emphasize discriminative cues, and underperform when facing diverse manipulation artifacts or domain shifts [6].

Advanced fusion approaches employ attention mechanisms for dynamic integration. Cross-attention, in particular, allows one modality (e.g., ViT embeddings as queries) to attend to the other (e.g., CNN features as keys/values), or bidirectional, facilitating context-aware feature interaction [8]. Such mechanisms reduce modality-specific biases, enhance discriminability, and improve detection of subtle deepfake artifacts, as demonstrated in cross-attention hybrids for vision tasks [8, 11].

Compared to static fusion in ViXNet [6], the proposed Cross-Attention Fusion (CAF) in CMAViT introduces explicit inter-branch attention. This adaptive strategy enables CNN local features to inform ViT global reasoning, yielding richer representations and superior performance over conventional fusion, especially in cross-dataset evaluation.

E. Cross-Dataset Generalization and Domain Shift

Cross-dataset generalization remains a fundamental challenge in deepfake detection, as models trained on a specific source dataset often suffer substantial performance degradation when evaluated on unseen target distributions. This degradation primarily stems from domain shift caused by differences in manipulation pipelines, compression levels, resolution, and post-processing artifacts. Conventional CNN-based detectors, while achieving strong intra-dataset performance, tend to overfit dataset-specific local artifacts, resulting in limited transferability across domains [3, 4]. Vision Transformer (ViT)–based architectures, due to their global self-attention mechanism, are less biased toward strictly local cues; however, when employed in isolation, they still remain vulnerable to domain shift in the absence of explicit feature fusion or adaptation strategies [5].

Hybrid CNN–Transformer models, such as ViXNet-style architectures[6] partially alleviate this limitation by combining local artifact sensitivity with global contextual reasoning. Nevertheless, existing fusion strategies based on feature concatenation or static attention are often insufficient to fully exploit

the complementary nature of CNN and ViT representations under cross-dataset settings.

In contrast, the proposed CMAViT explicitly addresses domain shift through a cross-attention fusion mechanism, enabling dynamic interaction between CNN-derived local features and ViT-based global representations. Unlike static fusion, cross-attention allows the model to adaptively reweight features depending on their relevance across domains, thereby reducing reliance on dataset-specific artifacts.

Following a strict inter-dataset evaluation protocol, CMAViT is trained on source datasets (OpenForensics and FaceForensics++) and evaluated on the unseen Celeb-DF (v2) dataset. Under identical training and evaluation settings, CMAViT consistently outperforms ViXNet-style fusion baselines in terms of AUC, demonstrating improved robustness to domain shift in both intra- and inter-dataset experiments. These results indicate that cross-attention–based feature interaction plays a critical role in enhancing cross-dataset generalization for deepfake detection.

III. PROPOSED METHOD

In this section, we present the proposed Cross-Modal Attention ViT–CNN Hybrid Network (CMAViT), an advanced extension of the ViXNet architecture [6] that improves deepfake detection robustness and cross-dataset generalization. The key contribution is the replacement of ViXNet’s simple feature concatenation with a cross-attention fusion (CAF) mechanism, enabling explicit and adaptive interaction between local CNN features (from Xception) and global token representations (from Vision Transformer). This attention-guided approach draws inspiration from recent hybrid models that leverage cross-attention to fuse complementary modalities in vision tasks.

Unlike the original ViXNet which relies on feature concatenation, our model introduces a token-level cross-attention mechanism that allows CNN-derived spatial tokens to attend ViT patch tokens.

A. Overall Architecture

The CMAViT model consists of four main components:

- **Xception Token Branch:** Extracts spatial feature tokens from a pretrained Xception network.
- **ViT Token Branch:** Extracts the full sequence of patch and [CLS] tokens from a pretrained Vision Transformer.
- **Cross-Attention Fusion Module:** Performs one-way cross-attention where Xception tokens attend to ViT tokens, followed by

pooled feature concatenation and MLP refinement.

- **Classification Head:** A lightweight MLP that produces binary real/fake logits.

The input is a preprocessed RGB image of size 224×224 . The model leverages pretrained backbones from the timm library. Unlike the original ViXNet, which concatenates globally pooled feature vectors, CMAViT operates at the token level to preserve spatial and patch-wise information during fusion [8].

B. Feature Extraction Branches

Xception Token Branch Xception, pretrained on ImageNet, is used with global pooling disabled to retain the final convolutional feature map (typically $2048 \text{ channels} \times 7 \times 7$ spatial resolution). This map is flattened and transposed to produce spatial tokens: 49 tokens $\times$ 2048 dimensions. These tokens capture rich local texture and manipulation artifacts [3].

Vision Transformer Token Branch A pretrained ViT (default: ViT-Base-Patch16-224) with global pooling disabled yields the complete token sequence: 197 tokens $\times$ 768 dimensions (including the [CLS] token and 196 patch tokens). This sequence encodes long-range dependencies and global semantic inconsistencies [5].

Both branches are initialized with pretrained weights.

C. Cross-Attention Fusion Module

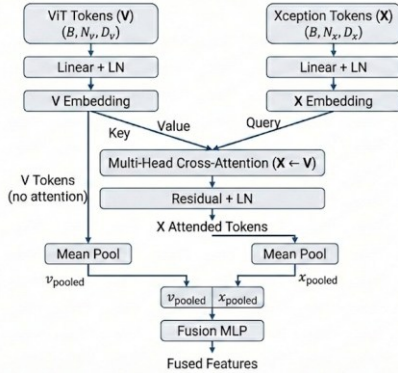


Figure 1. One-way cross-attention fusion module ($X \leftarrow V$), where Xception tokens attend to ViT tokens without reciprocal attention, enabling effective integration of global spatial and patch-level representations.

As illustrated in Figure 1, the proposed fusion module employs a one-way cross-attention mechanism, where Xception tokens query ViT tokens to selectively aggregate complementary information.

- ViT tokens and Xception tokens $V, X \in \mathbb{R}^{B \times N \times D}$ are linearly projected into a shared embedding space with dimension $D = 512$, followed by layer normalization.
- **One-Way Multi-Head Cross-Attention** Multi head cross-attention is applied with

Xception tokens as queries and ViT tokens as keys and values. The attention output is combined with the original Xception tokens via a residual connection and layer normalization, while ViT tokens remain unchanged [12].

- **Global Pooling and Fusion**

The attended Xception tokens and ViT tokens are separately aggregated using mean pooling, concatenated, and passed through a two-layer fusion MLP with ReLU activation and dropout to produce the final fused representation.

This attention-guided fusion improves robustness and cross-dataset generalization compared to static feature concatenation [13].

D. Classification Head and Full Forward Pass

The classification head is a multi-layer perceptron: Linear (fused_dim $\rightarrow$ 256) $\rightarrow$ Batch Normalization $\rightarrow$ ReLU $\rightarrow$ Dropout (0.5) $\rightarrow$ Linear (256 $\rightarrow$ 2). The forward pass involves extracting tokens from both branches, fusing them via cross-attention, and classifying the fused features.

E. Three-Stage Training Strategy

Training the proposed CMAViT model involves two large pretrained backbones (Xception and Vision Transformer) coupled through a cross-attention fusion module. Direct end-to-end fine-tuning of all components was found to be unstable and prone to overfitting. To address this issue, we adopt a three-stage progressive training strategy, which has proven effective in transfer learning and hybrid architectures [14, 15]. Across all stages, the model is optimized using the cross-entropy loss, with labels defined as Fake = 0 and Real = 1.

Stage 1: Fusion and Classification Head Training.

In the first stage, both pretrained backbones are frozen, and only the cross-attention fusion module and the classification head are trained. This allows the model to learn task-specific decision boundaries in the fused feature space while preserving the general visual representations learned from large-scale pretraining. Such head-only training is a common practice to stabilize optimization when adapting pretrained models to new tasks [15].

Stage 2: Partial Fine-Tuning of the Xception Branch.

In the second stage, the upper layers of the Xception token branch are partially unfrozen, while the Vision Transformer branch remains frozen. This enables the convolutional backbone to adapt local texture and spatial features to manipulation-specific artifacts commonly observed in deepfake images, while maintaining stable global representations. Different learning rates are applied to backbone and

fusion layers to prevent catastrophic forgetting, following standard gradual fine-tuning strategies [14].
Stage 3: Partial Fine-Tuning of the Vision Transformer Branch.

In the final stage, the last Transformer encoder blocks of the Vision Transformer token branch are unfrozen. This stage allows the model to refine high-level semantic representations and long-range dependencies, which are critical for detecting subtle and globally inconsistent manipulation patterns. A lower learning rate is used for the Transformer parameters due to their high capacity and sensitivity to overfitting.

After completing all three stages, the best-performing model is selected based on validation performance and evaluated on the test set using accuracy, precision, recall, and F1-score. This progressive training strategy ensures stable convergence and improves robustness and cross-dataset generalization of the proposed CMAViT model.

IV. EXPERIMENTAL SETUP

A. Datasets

Experiments are conducted on a unified dataset constructed from OpenForensics and FaceForensics++ (FF++) [9], with additional cross-dataset evaluation on Celeb-DF. In both FaceForensics++ and Celeb-DF, images are obtained by extracting frames from manipulated and pristine videos, which preserves realistic temporal artifacts and compression effects commonly observed in real-world deepfake scenarios.

OpenForensics provides diverse real and manipulated facial images collected from unconstrained sources, while FaceForensics++ serves as a standard benchmark with multiple state-of-the-art manipulation methods. Samples from OpenForensics and FaceForensics++ are merged and randomly split into training, validation, and test sets using a 70% / 15% / 15% ratio with balanced class distributions. The training set contains 88,531 fake and 85,799 real images; the validation and test sets each include 19,108 fake and 18,660 real images.

For cross-dataset generalization evaluation, an additional 10,000 frame-level images extracted from Celeb-DF videos are used exclusively for testing, without any overlap with the training data. All images are resized to 224x224 converted to RGB format and normalized using ImageNet statistics to match the pretrained Xception and Vision Transformer backbones.

B. Implementation Details

The proposed CMAViT model is implemented in PyTorch, using ImageNet-pretrained backbones from

the *timm* library. The architecture comprises an Xception-based CNN branch and a ViT-Base-Patch16-224 branch. For cross-attention fusion, global pooling is disabled in the Xception branch to preserve spatial feature maps, which are reshaped into token sequences, while the ViT branch retains the full token set, including the class token.[11].

Cross-modal interaction is realized via a one-way multi-head cross-attention mechanism, where Xception tokens attend to ViT tokens after linear projection into a shared 512-dimensional embedding space [8]. To ensure numerical stability under automatic mixed precision (AMP) training, cross-attention is computed explicitly in full precision (FP32). The attended tokens are mean-pooled and refined by a lightweight MLP before classification.

Training follows a three-stage progressive strategy, gradually unfreezing the Xception and ViT backbones. Optimization is performed using AdamW with $\beta_1 = 0.9$, $\beta_2 = 0.999$ and $\epsilon = 1 \times 10^{-8}$. Strong data augmentation—including JPEG compression, random downscaling, motion blur, Gaussian noise, and random erasing—is applied to improve robustness against real-world deepfake artifacts. All images are resized to 224x224 and normalized using ImageNet statistics.

C. Evaluation Metrics

To evaluate the performance of the proposed model, we adopt the Area Under the Receiver Operating Characteristic Curve (AUC) as the primary evaluation metric, together with the F1-score as a complementary measure. AUC provides a threshold-independent assessment of the model's discriminative capability, which is particularly suitable for deepfake detection under class imbalance. In addition, the F1-score reflects the balance between precision and recall at a fixed decision threshold, offering complementary insight into practical classification performance [2, 9].

V. RESULTS AND DISCUSSION

A. Intra-dataset Evaluation

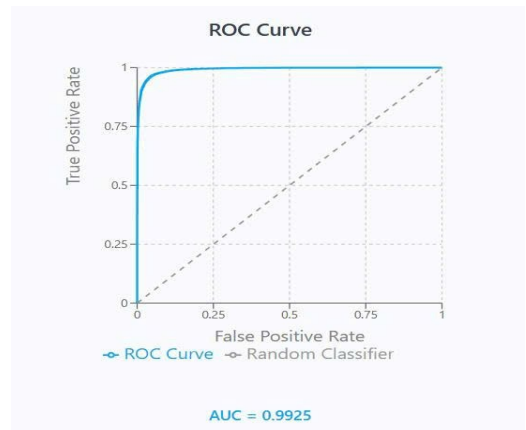


Figure 2. Receiver Operating Characteristic (ROC) curve of the proposed model on the test set. The curve illustrates the trade-off

between the True Positive Rate (TPR) and the False Positive Rate (FPR) across different decision thresholds.

We first evaluate the proposed ViXNet with Cross-Attention Fusion under the intra-dataset setting, where training and testing are conducted on the same data distribution. As shown in Fig. 2, the model achieves an AUC of 0.9925, along with 95.56% accuracy and 95.56% F1-score on the test set consisting of 37,417 samples.

Confusion Matrix		
Actual	Predicted	
	Fake	Real
	Fake	Real
Fake	17902	510
Real	1151	17854

Figure 3. Confusion matrix of the proposed model on the test set. The matrix summarizes the classification results for real and fake images, showing that the model correctly classifies most samples in both classes, with a low number of misclassifications.

B. Inter-Dataset Evaluation

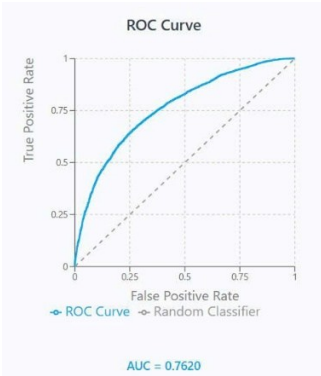


Figure 4. Receiver Operating Characteristic (ROC) curve of the proposed model under inter-dataset evaluation. The model

achieves an AUC of 0.7620, indicating its ability to generalize to unseen datasets despite distribution shifts between training and testing datasets.

Confusion Matrix		
Actual	Predicted	
	Fake	Real
	Fake	Real
Fake	3791	1245
Real	1835	3232

Figure 5. Confusion matrix of the proposed model under inter-dataset evaluation. The matrix summarizes the classification performance when training and testing are conducted on different datasets.

To further assess the generalization ability of the proposed model, we conduct inter-dataset evaluation, where the model trained on the source dataset is directly tested on a different target dataset without any fine-tuning. Under this more challenging setting, the model achieves an AUC of 0.7620, with 69.51% accuracy and an F1-score of 67.73% over 10,103 test samples.

Although there is a noticeable performance drop compared to the intra-dataset scenario, the results remain competitive given the substantial domain gap between datasets, including differences in compression level, identity distribution, manipulation methods, and post-processing artifacts. The confusion matrix reveals increased misclassification rates, particularly for real samples incorrectly predicted as fake, which is a common issue in cross-dataset deepfake detection.

C. Discussion

Table 1. Performance comparison under the OpenForensics+FF++_OpenForensics+FF++_CeDF protocol

Method	Training data	Accuracy (%)	AUC (%)
Xception [3]	FF++	55.60	58.06
F3-Net [16]	FF++	68.04	66.12
AMTEN [17]	FF++	44.21	50.96
CCViT [18]	FF++	63.89	53.89
ViXNet [6]	FF++	69.30	74.78
CMAViT (ours)	OpenForensics + FF++	69.51	76.20

Note: Celeb-DF is used exclusively for testing and is not involved in training or validation.

This section analyzes the performance of the proposed method in comparison with representative state-of-the-art deepfake detection approaches, with a particular focus on cross-dataset generalization. As discussed in the experimental setup, the model is trained using a combination of OpenForensics and FaceForensics++ and evaluated on Celeb-DF, which is not seen during training. This setting is intentionally designed to assess the robustness of the learned representations under domain shift, a known challenge in deepfake detection.

Table 1 summarizes the quantitative comparison under this inter-dataset protocol. All methods are evaluated on the same Celeb-DF test set, ensuring a fair comparison in terms of generalization performance. The reported results highlight the effectiveness of the proposed approach in leveraging multi-source training data to improve cross-dataset robustness, while avoiding any exposure to target-domain samples during training.

Table 1 presents the cross-dataset evaluation on the unseen Celeb-DF dataset. CMAViT achieves the highest performance with 76.20% AUC, significantly outperforming single-source CNN baselines (Xception [3]) and frequency-based methods (F3-Net [16], AMTEN [17]), which suffer from domain overfitting. Notably, our approach surpasses other hybrid architectures like ViXNet [6] and CCViT [18]. This demonstrates that combining multi-source training (OpenForensics + FF++) with dynamic cross-attention fusion yields superior generalization compared to static concatenation strategies, effectively mitigating the domain shift between training and testing distributions.

VI. CONCLUSION

This paper presents CMAViT, a hybrid architecture that integrates local CNN features with global ViT contexts via a cross-attention mechanism. Unlike static concatenation, our token-level fusion enables dynamic interaction between modalities, significantly improving robustness under cross-dataset evaluation, as evidenced by superior performance on Celeb-DF. Additionally, the progressive three-stage training strategy effectively stabilizes the optimization of pretrained backbones. Future work will extend this framework to temporal modeling for video deepfake detection and optimize computational efficiency for real-world deployment.

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